

OS Unit 3 — Deadlocks Complete Notes

Operating Systems (OS) · Unit 3 · Deadlocks · 2025

Overview

These notes cover **Deadlocks**. How operating systems manage processes, CPU time, memory, storage and devices — and keep it all safe and concurrent. Each section states the core idea, a worked intuition and the points examiners look for.

1. Deadlock Characterization

Deadlock requires mutual exclusion, hold and wait, no preemption and circular wait simultaneously.

Why it matters: Deadlock Characterization appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of deadlock characterization in one line.
- A small example showing deadlock characterization in practice.
- Common mistakes students make and how to avoid them.

2. Resource Allocation Graph

A cycle in a single-instance resource allocation graph implies deadlock.

Why it matters: Resource Allocation Graph appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of resource allocation graph in one line.
- A small example showing resource allocation graph in practice.
- Common mistakes students make and how to avoid them.

3. Deadlock Prevention

Prevention denies at least one of the four necessary conditions, e.g. by ordering resources.

Why it matters: Deadlock Prevention appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of deadlock prevention in one line.
- A small example showing deadlock prevention in practice.
- Common mistakes students make and how to avoid them.

4. Deadlock Avoidance

Avoidance keeps the system in a safe state by checking each request before granting it.

Why it matters: Deadlock Avoidance appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of deadlock avoidance in one line.
- A small example showing deadlock avoidance in practice.
- Common mistakes students make and how to avoid them.

5. Banker's Algorithm

The Banker's algorithm uses Available, Max, Allocation and Need matrices to find a safe sequence.

Why it matters: Banker's Algorithm appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of banker's algorithm in one line.
- A small example showing banker's algorithm in practice.
- Common mistakes students make and how to avoid them.

6. Deadlock Detection and Recovery

Detection algorithms find deadlocked processes; recovery terminates processes or preempts resources.

Why it matters: Deadlock Detection and Recovery appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of deadlock detection and recovery in one line.
- A small example showing deadlock detection and recovery in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Deadlock Characterization · Resource Allocation Graph · Deadlock Prevention · Deadlock Avoidance · Banker's Algorithm · Deadlock Detection and Recovery